

Plan: `cycles.rs` Strategy Refactor

Collapsing 16 Cartesian PyO3 Wrappers into 2 + `EnumerationConfig`

2026-05-11

1 Scope

`hymeko_py/src/cycles.rs` currently exposes **16 distinct** `#[pyfunction]` enumerator variants whose names encode a Cartesian product over axes {enumeration mode, scorer, pruner, vertex-filter, ABB strategy, batched output, tiered caps}. Average wrapper is 134 LOC, mostly copy-paste boilerplate that re-implements scorer dispatch, pruner dispatch, filter construction, SoA conversion, and PyArray output. Total 2251 LOC, 0 structs, 0 traits.

The underlying Rust algorithm crate `hymeko_graph` *already* provides clean Strategy traits and impls (`Scorer`, `Pruner`, `VertexFilter`, with 4–5 concrete implementations each). This refactor preserves those traits and replaces the 16 wrappers with **two** family entry points plus one configuration struct. No algorithm change.

2 Goal

- One `#[pyfunction]` per *family* (cycles, walks) — not per Cartesian cell.
- All toggles enter through `EnumerationConfig` (`score_kind`, `pruner_kind`, `filter_kind`, `abb_mode`, `batched`, `tiered`, ...).
- Internal dispatcher matches (`mode`, `abb`, `batched`, `tiered`) and delegates to the existing parameterized algorithm functions in `hymeko_graph::topk_cycles`.

3 Affected files

- `hymeko_py/src/cycles.rs` — full rewrite of the PyO3 surface. Internal helpers preserved. Expected: 2251 → ~800 LOC.
- `hymeko_py/src/lib.rs` — `m.add_function` registrations updated to point at new entry-point names.
- **24 Python call sites** (counted via `grep`) across `signedkan_wip/` that currently use the legacy names — migrated to the new `enumerate_cycles_rs(..., config)` signature. The most touched callers in recent work: `signedkan_wip/src/run_gomb_smoke.py`, `signedkan_wip/src/cycle_cache`, `signedkan_wip/src/n_tuples.py`.

CORE.YAML items touched: none. `hymeko_py` is not listed in `Core.yaml` (which protects only the not-yet-existing `crates/hymeko_*` paths). Refactor permitted without escalation per `CLAUDE.md` §1.

4 Public API

```
# New entry points (Python-facing)
enumerate_cycles_rs(
    edges_u, edges_v, edges_s, n_nodes,
    k_len, m_per_vertex, # m_per_vertex=0 means "all-mode"
    score_kind="fraction_negative",
    pruner_kind="none",
    filter_kind="none",
    filter_min_degree=2,
    abb_mode="none", # "none" / "start_local" / "global_min"
    batched=True,
    tiered=False,
    fullness_gate=0.25,
    adaptive_m_v=None,
) -> tuple[ndarray, ndarray] # (cycles (M, k), scores (M,))

enumerate_walks_rs(
    edges_u, edges_v, edges_s, n_nodes,
    walk_len, m_per_vertex,
    score_kind="fraction_negative",
    ...
) -> tuple[ndarray, ndarray]
```

The 16 legacy names (`enumerate_top_k_per_vertex_cycles_signed_filtered_batched_rs, ...`) are **removed**. Callers migrate to the new signature in the same commit — no shim period (per memory `feedback_decompose_long_files`: clean breaks beat deprecation cruft).

5 Internal structure (Rust)

```
#[derive(Debug, Clone)]
pub struct EnumerationConfig<'a> {
    pub k_len: usize,
    pub m_per_vertex: u32, // 0 = all-mode
    pub score_kind: &'a str,
    pub pruner_kind: &'a str,
    pub filter_kind: &'a str,
    pub filter_min_degree: u32,
    pub abb_mode: AbbMode, // None | StartLocal | GlobalMin
    pub batched: bool,
    pub tiered: bool,
    pub fullness_gate: f64,
}

fn dispatch_cycle_enum(
    g: &SignedGraph, cfg: &EnumerationConfig,
) -> Result<Batch, EnumError> {
    let scorer = pick_scorer(cfg.score_kind)?;
    let pruner = pick_pruner(cfg.pruner_kind)?;
    let filter = build_vertex_filter(cfg.filter_kind, cfg.filter_min_degree)?;
    let keep = filter.keep_set(g);
    match (cfg.m_per_vertex, cfg.abb_mode, cfg.batched, cfg.tiered) {
        (0, _, _, _) => /* all-mode */,
        (m, AbbMode::None, true, false) => /* per-vertex batched */,
        (m, AbbMode::StartLocal, true, false) => /* per-vertex BB batched */,
        (m, AbbMode::GlobalMin, true, false) => /* per-vertex BB-global batched */,
    }
```

```

        (m, _, _, true)                => /* tiered cap variant */,
        ...
    }
}

#[pyfunction]
pub fn enumerate_cycles_rs(...) -> PyResult<(...)> {
    let g = build_signed_graph(...)?;
    let cfg = EnumerationConfig { ... };
    let batch = py.detach(|| dispatch_cycle_enum(&g, &cfg));
    batch_to_pyarray(py, batch?, cfg.k_len)
}

```

6 Test strategy

- **Rust:** `cargo test -p hymeko_py` continues to pass. Add 1 dispatch-table test: every (mode, abb, batched, tiered, scorer, pruner) combination produces a non-empty result on a small fixture or returns a typed error.
- **Python:** all existing `signedkan.wip/tests/` pass after migration (currently the suite passes; baseline is the 13 + 15 + ... gomb/etc. tests we ran today plus the cycle.cache tests at `signedkan.wip/tests/`
- **Regression:** compare cycle output (sorted) between the old and new wrappers for at least one (scorer × pruner × filter) cell each on Bitcoin OTC, before deleting the legacy pyfunctions. Output equality (modulo ordering) is required.

7 Performance budget

No algorithm change. Expected wall-time delta: $\pm 5\%$ noise (single extra match-on-strings per call; negligible vs. DFS enumeration). RSS unchanged. If we measure $> 10\%$ regression, CLAUDE.md §3 forces investigation per the symmetric rule.

8 Risk anticipation

- **Caller-side breakage:** 24 Python call sites need migration. Mitigation: refactor in one commit; smoke-run `run_gomb_smoke.py` on Bitcoin OTC after migration to catch typos.
- **Cycle output ordering:** the per-vertex enumerator returns cycles in some implementation-defined order. Sorted-set equality required for the regression test, not list equality.
- **Cache fingerprint drift** (memory `feedback_cycle_cache_fingerprint`): `cycle_cache.py`'s `_topk_fingerprint` dict must include every config field that affects the output. Refactor must preserve fingerprint coverage — i.e. add `abb_mode`, `tiered`, `batched` to the fingerprint if they weren't already (likely missing — that was the canonical failure mode in the prior memory).
- **ABB lossy variants:** the `_bb_` variants produce a SUBSET of the non-BB output (UB-pruning by construction). Regression test must use the same `abb_mode` on both sides, not compare across modes.

9 Rollback

`git revert <commit>`. Single commit per CLAUDE.md §0 checkpoint discipline.

10 Sequencing

1. Write the plan (**this doc**).
2. Introduce `EnumerationConfig` + `AbbMode` + `dispatch_cycle_enum` alongside the existing wrappers (zero-impact additive change).
3. Add the 2 new `#[pyfunction]`s (`enumerate_cycles_rs`, `enumerate_walks_rs`) calling the new dispatcher.
4. Regression test: confirm new wrappers produce identical (sorted) output to the legacy ones on Bitcoin OTC for at least one cell per axis.
5. Migrate all 24 Python call sites to the new signature.
6. Run full test suite (Rust + Python).
7. Delete the 16 legacy `#[pyfunction]`s and their `lib.rs` registrations.
8. Re-run `cargo clippy --all-targets -- -D warnings` and `pytest -p no:randomly` as gates per CLAUDE.md §6.3.
9. Write the report (`reports/2026-05-11-cycles-strategy-refactor.md`).

11 Acceptance gates

- All Rust tests pass.
- All Python tests pass.
- `cargo clippy` clean under `-D warnings`.
- `wc -l hymeko_py/src/cycles.rs < 1000` (target 700–900).
- Smoke run: Bitcoin OTC gomb 5-seed (existing `reports/2026-05-11-hymeko-gomb-sphere.md` numbers) reproduces within ± 0.001 AUC.
- Old pyfunction names are gone (`grep -c "enumerate_top_k" hymeko_py/src/cycles.rs` returns 0 or 1, not 16).